

Question Bank Numerical Questions (6-10 Marks Each)

Evaluation Metrics – Real-Life Based

Q1: Medical Diagnosis System

A hospital develops a model to detect a serious disease. The results are:

- True Positives (TP) = 120
- False Positives (FP) = 30
- False Negatives (FN) = 20
- True Negatives (TN) = 230

Solve the following -

1. Calculate **Accuracy**
2. Calculate **Precision**
3. Calculate **Recall**
4. Calculate **F1 Score**
5. Comment on model suitability for medical diagnosis

Q2: Spam Email Classification

An email filtering system produced the following results:

- TP = 200 (Spam correctly identified)
- FP = 50 (Normal emails marked as spam)
- FN = 40 (Spam missed)
- TN = 710 (Correctly identified normal emails)

Solve the following -

1. Compute **Accuracy, Precision, Recall, F1 Score**
2. Analyze whether the model is reliable
3. Suggest which metric should be improved and why

Q3: Credit Card Fraud Detection

A bank uses a fraud detection system:

- TP = 40
- FP = 10
- FN = 60
- TN = 890

Solve the following -

1. Calculate all evaluation metrics
2. Is accuracy a good measure here? Justify
3. Which metric is most important in fraud detection and why?

Q4: Student Performance Prediction

A university predicts whether students will pass or fail:

- TP = 150
- FP = 30
- FN = 20
- TN = 100

Solve the following -

1. Calculate **Accuracy, Precision, Recall, F1 Score**
2. Interpret the results
3. Is the model balanced? Explain

Q5: COVID-19 Screening System

A screening model gives:

- TP = 300
- FP = 100
- FN = 50
- TN = 550

Solve the following -

1. Calculate all performance metrics
2. Analyze the impact of false negatives
3. Which metric should be prioritized in pandemic situations?

Hint Steps to write answer of above questions in SPPU exam

For 6-10 marks, write:

- Step 1: Confusion Matrix values
- Step 2: All formulas
- Step 3: Calculations
- Step 4: Final answers
- Step 5: Interpretation (very important for full marks)